

Question	Marking guidance	Mark	Comments
02.1	X-rays can produce more detail OR clearer image OR better resolution.	1	Ignore comments about contrast
02.2	Ultrasound is non-ionising Link to risk of cancer or mutations	1	Accept real-time or moving image for 1 mark
		1	Accept damage to tissue / cell / cell nuclei

Question	Marking guidance	Mark	Comments
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10	D
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22	A
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23	B
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Question	Marking guidance	Mark	Comments
01	description of longitudinal wave ✓ description of polarisation OR only transverse waves can be polarised ✓	2	eg oscillations/vibrations occur parallel to direction of energy/wave eg restriction of oscillations/vibrations to one plane/direction

Question	Marking guidance	Mark	Comments
01	Use of $\lambda = \frac{h}{p}$ or $\lambda = \frac{h}{mv}$ ✓ 5.64×10^{-10} (m) ✓	2	When substituting into $\lambda = \frac{h}{p}$ do not accept v instead of p Accept 5.6×10^{-10} (m)

	$f = 4.6(4) \times 10^{14}(\text{Hz})$ ✓		
03	rotate the filter ✓ correct set-up described or idea that it is necessary to measure, monitor or observe (the variation in) intensity during rotation ✓ if the intensity varies it is polarised and/or if the intensity does not vary, it is not polarised or if it is polarised, when rotated, a minimum is observed ✓	3	Correct setup could be on a diagram For correct setup, must be clear that the filter is between the lamp and the meter ignore any reference to the filter reducing the intensity of the light as it would do this in either case do not award MP3 for answers involving discrete measurements only

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
04.1	any mention of resonance ✓ the idea that there is a periodic driving force or a forcing oscillation ✓ (Max amplitude or resonance when) the frequency of the forcing oscillation matches the resonant/natural frequency of vibration (of the machine) ✓	3	Look for the idea that there is a force causing the oscillation and that the applied force is periodic or oscillating Not just the rotation of the drum for MP2
04.2	smooth curve with a single peak ✓	1	
04.3	damping reduces the amplitude or peak is lowered ✓ idea that damping removes energy from the system ✓	1 1	Ignore reference to the width of the peak or its position on the graph

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
07.1	(as the current increases,) the voltage across the internal resistance increases ✓ so less of the total voltage is available to the external resistance or correct equation discussed eg terminal pd = $E - Ir$ ✓ Alternative R decreased ✓ R has a smaller share of the voltage wtte ✓	2	allow 'lost volts increase' for MP1 must be clear which resistor is being discussed For MP2, allow 'terminal pd decreases' Must include that E is fixed but condone omission of statement that r is fixed must be clear which resistor is being discussed
07.2	line of best fit drawn <u>and</u> intercept used ✓ 5.7–5.9 V ✓	2	Expect to see an additional line for 07.4

	attempt to take the gradient ✓ correct readings, with evidence seen on the graph defining 2 points at least 2.5 A apart ✓ 1.15–1.27 (Ω) ✓ Alternative: use of their value of E and a point on the line of I greater than 2.5 A. ✓ Use of $V = E - Ir$ ✓ 1.15–1.27 (Ω) ✓		must be from the line and not from two data points for both MP1 and MP2 Accept use of both intercepts without annotation of graph reject a negative number for MP3 Expect to see triangle with greater than half size of graph.
07.4	straight line with negative gradient and V intercept of 2.7 to 3.0 ✓ straight line with negative gradient and half the gradient of the original ✓	2	MP1 and MP2 can be given even when no best fit line has been drawn for 07.1/07.2. Line can be extrapolated to judge intercept Gradients judged by eye note: it will have the same I intercept if both marks are correct

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Additional comments/guidelines	Mark
05.1	clear rearrangement including: squaring f correctly ✓ substituting m/l for μ ✓		2
05.2	gradient determined correctly = 17.6 to 18.2 ($\text{Hz}^2 \text{N}^{-1}$) ✓ correct substitution and rearrangement of $1/(4lm)$ ✓ $m = 5.3 \times 10^{-3} \text{ (kg)}$ to $5.5 \times 10^{-3} \text{ (kg)}$ ✓	For MP1, evidence of gradient calculation must be seen OR if a single point rather than a gradient is used, evidence of extending the line to the origin must be seen. must be at least 2 sf in final answer	3
05.3	smaller uncertainty for the tape measure ✓ because the metre rule must be used multiple times ✓	MP1 condone smaller error or more <u>precise</u> for smaller uncertainty	2

Question	Marking guidance	Additional comments/guidelines	Mark
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Question	Answers	Additional comments/Guidelines	Mark	ID details
02	(Idea that reflection at ends produces) two waves (travelling) in opposite directions ✓ Waves have the same frequency/wavelength (and similar amplitude) ✓ Idea that the amplitude of the stationary wave is the (algebraic) sum of the original wave and its reflection ✓	If there is a suggestion that the waves are in phase or coherent allow MAX 2 Accept mention of superposition of the two waves for MP3 Condone superimpose for superpose	3	E
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	ID details
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Question	Answers	Additional comments/Guidelines	Mark	AO
04	Physical evidence for the existence of energy levels✓ Explanation of how energy levels give rise to photons/quanta of particular energies/frequencies✓ Discussion of why the energy levels must be discrete ✓	E.g. for MP1: line (emission and/or absorption) spectra Allow 'not continuous' for line ✓ e.g. for MP3 why spectra have a limited number of frequencies or wavelengths Accept similar argument for characteristic spectrum of X-rays	3	3 × AO1
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Marking guidance	Mark	Comments
08.1	Same frequency ✓ Constant phase relationship/difference ✓	2	Do not accept 'in phase'

Question	Marking guidance	Mark	Comments
08.2	MAX 3 Waves (from the loudspeakers) in anti-phase/180° out of phase/π (rad) out of phase at B ✓ Path difference at B = $\lambda/2$ or $3\lambda/2$...etc✓ Reference to superposition ✓ Destructive interference occurs ✓	3	Treat any discussion of process at A as neutral Do not accept path difference in MP1 or phase difference in MP2 Condone superimpose for superpose

Question	Marking guidance	Mark	Comments
08.4	(Increasing frequency decreases wavelength and therefore) At A there will be no change / at A there will (still) be a maximum as path difference (still) zero/waves (still) in phase.✓ At B the intensity will increase as the waves no longer in antiphase/ no longer cancel out/ path difference no longer half a wavelength.✓	2	For MP1 allow intensity at A does not change as it is the central maximum. For MP2 condone argument based on change in w

Question	Marking guidance	Mark	Comments
08.5	Idea that the sound is reflected from the walls ✓ (and) that this will either increase the intensity at B (by constructive interference) OR reduce the intensity at A (by destructive interference) ✓	2	Accept: will affect the intensity due to constructive/destructive interference without reference to A or B

Total		12
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Question	Marking guidance	Mark	Comments
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15	c	$0.82f$
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17	D	are reflected well at tissue boundaries.
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18	B	$0.29T$
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19	A	$t = 0.10 \text{ s}$
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	Involve oscillations/vibration (about mean equilibrium position) at right angles ✓ to direction of energy transfer ✓	Accept displacement of vibrating particles at right angles MP1 must have idea of oscillation/vibration and at right angles. Ignore references to polarisation	2	2 × AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	A material whose resistivity/resistance is zero ✓ below the <u>critical</u> temperature ✓	Condone 'negligible' for 'zero'	2	2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Can generate strong magnetic fields OR They can sustain large currents ✓ There are reduced thermal/energy losses OR No damage due to overheating (of the electromagnets) ✓	Accept 'no' for 'reduced'	2	2 × AO1
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04	Hold filter in front of sunglasses and rotate (through 360°) ✓ Any variation of intensity during rotation means polarising OR No variation during rotation means not polarising ✓ Reduced intensity occurs when polarisation planes are not aligned OR minimum when polarisation planes crossed ✓	If no other marks given, give 1 mark for either MP1 or MP2, without mention of the sunglasses.	3	3 × AO2
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	2 MAX from Idea of two waves travelling in opposite directions ✓ superpose /constructively interfere/give maximum amplitude ✓ because they are in phase ✓	Do not award MP3 if suggestion that waves are coherent.	2	2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Evidence of use of all 3 dots ✓ ($\lambda = 2$ measurements added together or one single measurement from dot one to dot three) Use of $c = f\lambda$ with correct substitution ✓ Answer consistent with their measurement ✓	Evidence can be on figure 4 or a written statement Allow POT error in MP2	3	1 × AO1 1 × AO3 1 × AO4
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	A	6.25×10^9	AO2
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19	C	The amplitude is a minimum at Q .	AO3
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23	A	4.0 25	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Max 2 from: ✓✓ different wavelengths take different times to travel (through core) (because) different wavelengths travel at different speeds (in core) (because) refractive index of fibre varies with wavelength	Do not allow suggestion of modal dispersion. Condone 'frequency' for 'wavelength'. 'Spread out more' is insufficient for 'different times'. Ignore any specific correlation between wavelength and wave speed.	2	2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	idea that increased length of fibre leads to: increased pulse broadening OR reduced intensity / amplitude ✓ idea that (max) transmission rate depends on: having distinct time gaps between (adjacent) pulses OR pulses having a minimum intensity ✓	MP1: Condone 'dispersion' will increase with length. Reference to dispersion can be for material or modal. MP1: Allow a specific reason for reduced intensity e.g. absorption / scattering / refraction (out of core) / attenuation. Allow 'weaker signal' for reduced intensity. For full credit their MP1 and MP2 must correlate.	2	2 × AO2
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	converts E to $8.32 \times 10^{-19} \text{ J}$ OR uses 5.2 and a value for h in $E = \frac{hc}{\lambda}$ or $E = hf$ ✓ $2.4 \times 10^{-7} \text{ (m)}$ ✓	Expect $f = 1.25 \times 10^{15} \text{ Hz}$ Calculator value is $2.3906 \times 10^{-7} \text{ (m)}$	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	idea that UV photon is absorbed by atom in the powder coating and is excited ✓ idea that atom (in the powder coating) de-excites and emits photons ✓ links de-excitation to emission of photons that have less energy (than UV and are visible) ✓	MP1: Condone photon “collides” with atom Allow 1 mark (for MP1 and MP2) for a description of excitation and de-excitation without reference to atoms. Condone “in smaller energy steps”. Accept longer wavelengths or smaller frequencies.	3	AO1
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	use of $f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$ ✓ $\mu = 2.1 \times 10^{-4} \text{ (kg m}^{-1}\text{)} \checkmark$ $2.5 \times \text{their } \mu \checkmark$	MP1: Full substitution or algebraic rearrangement with μ as subject. Condone substitution into incorrect algebraic rearrangement. MP2 Calculator value: $2.0828 \times 10^{-4} \text{ (kg m}^{-1}\text{)}$ Calculator value: $5.2071 \times 10^{-4} \text{ (kg)}$	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	330 (Hz) ✓	Expect to see 325 (Hz)	1	AO3
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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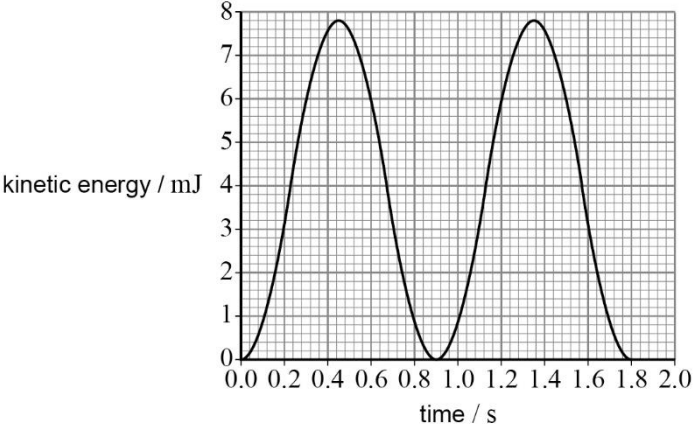
Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	amplitude is 1.8 mm ✓ 7.2 (mm) ✓	Allow MP1 for 3.6 mm	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	1.60×10^{-3} (s) ✓	Allow 2 sf answer	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	answer in range of -1.0 to -1.1 (mm) ✓	negative sign must be present	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	630 OR $\frac{1}{\text{their 04.2}}$ ✓	expect 625 (Hz)	1	AO1
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	correct general shape (2 peaks) ✓ both axes labelled e.g. “E_k / mJ” and “time / s” ✓ scales on each axis: E_k values from 0 to 7.8 and time values from 0 to 1.8 ✓	MP3: sketch must take up at least half of grid on each axis; sketch should not extend beyond 7.8 on their E_k axis or beyond 1.8 on their time axis; sketch should not correspond to negative value of E_k 	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	halves 1.80 to get 0.90 ✓ subtracts 0.90 or their half period from 1.61 ✓ doubles (1.61 – 0.9) to get 1.42 s ✓ OR $\frac{T_1}{2} + \frac{T_2}{2} = 1.61 \checkmark$ $T_1 + T_2 = 3.22 \checkmark$ $T_2 = 3.22 - 1.80 = 1.42 \text{ s } \checkmark$	Candidate may use alternatives to T_1 and T_2	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	uses 1.4 in $T = 2\pi \sqrt{\frac{l}{g}}$ ✓ $l = 0.487 \text{ m } \checkmark$ $x = 0.32 \text{ (m) } \checkmark$	MP1: Substitutes into equation or rearranges algebraically for l. Condone use of “x” for “l” in MP1 but not in MP2. Use of 1.42 s gives $l = 0.501 \text{ m}$ and $x = 0.304 \text{ m}$.	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	rate of energy transfer is greater between O and R because: amount of energy transfer is same (for Q to O as O to R) ✓ but duration of transfer is less ✓	Allow reverse arguments.	2	AO2
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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15	B	The frequency of the light does not change.	
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24	D	3.2 m	
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	$T = \frac{60}{39}$ OR $f = 39/60$ OR use of $T = 2\pi \sqrt{\frac{l}{g}}$ or equivalent with their T ✓ 0.59 (m) ✓	39/60 must clearly be a frequency Accept answers that round to 0.588 or 0.589 Reject 0.60	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	the natural/harmonic frequency of (the air in) the cylinder matches this (driving) frequency (of the loudspeaker) ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Stationary at P, maximum amplitude at Q ✓ Vertical (or parallel to PQ) oscillations at Q ✓	Ignore references to random motion of particles Condone vibrate for oscillate	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	$\lambda = 4(L + x)$ AND $v = f\lambda$ combined without errors in manipulation ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Correct read off from large triangle ✓ Gradient calculated and use of gradient = $\frac{4}{v}$ ✓ leading to 310–320 (m s⁻¹) ✓	Some annotation on the graph must be seen for MP1 Condone POT error for MP1 and MP2 MP2 may be combined into one step do not credit MP3 without any working 2sf or 3sf for MP3	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.5	Idea that it is the x-intercept OR idea that it is the y-intercept multiplied by $\frac{v}{4}$ ✓ description of how to determine the x- or y-intercept mathematically eg using similar triangles ✓ Alternative: Idea of one data point AND gradient/previously-calculated v being used ✓ and substituted into the equation $\frac{1}{f} = \frac{4L}{v} + \frac{4x}{v}$ ✓	Reject MP2 for a graphical extrapolation or just simply the word 'extrapolate'	2	AO4
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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19	D	2K	$\frac{1}{H}$		AO1

20	A	4.5λ	longitudinal	AO1
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21	B	457 Hz	AO1
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25	D	$\frac{1}{hc}$	AO1
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Total 14 marks

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	0.027 (m) ✓	Calculator value is 0.02727 m	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Links observed effects to polarisation ✓ Comments on effects for the two orientations of grille ✓ Links polarisation as a property of only transverse waves, so microwaves are transverse waves ✓	MP1: e.g. “the grille acts as a polariser” Any reference to grille causing polarising must refer to orientation of detector for further marks. MP2: e.g. “in Fig. 3 microwaves are blocked by grille; in Fig. 4 microwaves pass through grille” MP2: Condone incorrect correlation between orientation of grille and polarisation direction. MP3: Condone absence of word “only”.	3	1 × AO1 1 × AO2 1 × AO3
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Makes k the subject of period of mass–spring system OR calculates $T = 0.56$ (s) ✓ Allow a value that rounds 25 or 26 (N m^{-1}) ✓	MP1: Expect $T = 0.5555$ s MP2: Allow 1 mark for 2.4 N m^{-1} (from correct rearrangement for k but using 1.8 as T)	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Maximum amplitude occurs because resonance occurs (at 1.8 Hz) ✓ Maximum amplitude occurs when: frequency of forced oscillation equals frequency of free oscillation OR driving frequency equals natural/resonant frequency ✓	MP1 is for an explanation of why the amplitude has a maximum value MP2 is for an explanation of why the maximum amplitude occurs at 1.8 Hz MP2: Condone “applied” frequency.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Max 2: ✓✓ Amplitude of oscillations will be smaller at all frequencies or at resonance frequency/1.8 Hz or maximum amplitude will be smaller (with water/Fig. 9 compared to without water/Fig. 8) Maximum amplitude oscillation/resonance still occurs at 1.8 Hz or natural frequency is still 1.8 Hz Increased (rate of) energy dissipation (for the same amplitude/speed)	MP1: “there are smaller amplitude oscillations” is insufficient MP2: allow natural/resonant frequency may be (slightly) less than 1.8 Hz (due to damping)	2	AO2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Same frequency / wavelength ✓ Constant phase difference / relationship ✓	MP2: "In phase" is insufficient.	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Waves (from the loudspeakers) in anti-phase / 180° out of phase / π (rad) out of phase (at minimum) ✓ Path difference (at minimum) = $\frac{\lambda}{2}$ ✓ Reference to superposition or destructive interference occurring ✓	If no other mark given, allow 'wavelength is 20 cm' for 1 mark. Treat reference to "node" as neutral.	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	0.10 (m) with some relevant explanation ✓ Explanation in terms of interference and path difference OR in terms of stationary waves ✓	Allow 'wavelength is 20 cm' for 1 mark if not already credited in 08.2 MP1: Condone 1 sf MP2: e.g. minimum-to-minimum spacing is half wavelength; maximum-to-minimum is quarter wavelength Accept "node" for minimum, and "antinode" for maximum.	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Amplitude of wave from B > amplitude of wave from A ✓ Idea that superposition of the two unequal amplitudes produces a larger resultant amplitude (than previous minimum) ✓	MP2: Allow idea that, due to superposition, resultant amplitude cannot be zero or that a node cannot be produced. MP2: Allow idea that (magnitude of) destructive interference is reduced.	2	AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	B	down up	AO3
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16	B	$\frac{2\lambda D}{s}$	AO2
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17	C	$\frac{4\pi^2}{S}$	AO2
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18	A	decreases remains the same	AO1
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21	D	$\frac{9T}{4}$	AO2
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22	A	32°	AO1
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23	B	41 kJ	AO3
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25	A	$4.8 \times 10^{-20} \text{ J}$	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Reference to wave equation AND frequency stays the same ✓	Or reverse argument	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	$\left(\frac{\sin 59}{\sin 31} =\right) 1.4$ ✓	Reject if any units are given	1	AO2

Total			2	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	2.5 (s) ✓ Idea that there are two maxima/minima of energy/height in one period ✓		2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Use of $T = 2\pi\sqrt{\frac{l}{g}}$ ✓ 1.6 (m) ✓	ecf from 04.1 for both marks	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Any one from: ✓ - the motion is damped (by air resistance/friction) - the motion is driven (by the child kicking her legs) - the angle of oscillation is not small - the child is not a point mass / mass of chain is significant - the chain does not stay taut.		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Sinusoidal shape with correct period ✓ Decreasing peaks ✓ Minima touching x -axis at 0, 1.25, 2.5; maxima at approximately 0.63, 1.88 ✓		3	AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	(Plane) polarisation is when the direction of oscillation is restricted to a certain plane (perpendicular to the direction of propagation of the wave) ✓ The idea that there are no oscillations perpendicular to the direction of propagation in longitudinal waves OR only transverse waves have oscillations perpendicular to the direction of propagation ✓		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	When signals are polarised, the antenna/receiver must be aligned with the plane of the signal ✓ Idea that mobile phones can be moved around and held in different orientations OR that the signal quality would fluctuate (as the user moves) ✓		2	AO2

Total			4	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Any two from: ✓✓ • energy is transferred along a progressive wave, but stored in a stationary wave • the phase of the oscillations varies continuously along a progressive wave, but is constant between two adjacent nodes/antinodes in a stationary wave • stationary waves have nodes and antinodes, while progressive waves do not OR the peaks and troughs (or compressions, rarefactions) move in a progressive wave but don't in a stationary wave • a progressive wave has the same amplitude at all points whereas the amplitude varies along a stationary wave.		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	The mark scheme gives some guidance as to what statements are expected to be seen in a 1 or 2-mark (L1), 3 or 4-mark (L2) and 5 or 6-mark (L3) answer.		6	2 × AO1 4 × AO4
	Mark	Criteria		
	6	All three areas covered with at least two aspects covered in some detail. There should be at least one statement from the conclusions. Can be awarded even if there is an error and/or parts of one aspect missing.		
	5	A fair attempt to analyse all three areas, with two areas discussed successfully and one are partially.		
	4	Two areas successfully discussed, or one Discussed and two others covered partially. Whilst there will be several gaps, there should only be an occasional error.		
	3	One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.		
	2	Only one area discussed or makes a partial attempt at two areas.		
	1	None of the three areas covered without significant error.		
	0	No relevant comments		
		the measurements to be taken, including details of the measuring instruments used measure hanging mass with mass balance measure L with ruler or tape measure vary hanging mass OR L measure f for first harmonic (or second) using the signal generator techniques to reduce the uncertainty in the experiment at least 6 values of hanging mass OR L used repeat and average / remove anomalies how μ is determined using a graphical method valid combination of x and y variables algebraic manipulation of $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$ into the form $y = mx + c$ express μ in terms of the gradient Accept answers in terms of the 2nd harmonic		

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	Idea that the second harmonic has a node in the centre and it is easier to observe a node than an antinode ✓		1	AO4
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	B	1.2 <i>f</i>	AO1
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18	D	24 cm	AO1
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19	B	0.64 m	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Any one from: ✓ - the oscillation is damped - energy is removed from the system - air resistance OR friction at the point of suspension		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	X drawn vertically below the bob at equilibrium position ✓ Idea that the speed is greatest at this point OR shortest transit time ✓	Condone X drawn at least halfway down the string	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	Reduces the percentage uncertainty (in T) ✓ for the same absolute uncertainty over a longer time ✓	Allow relative, fractional for percentage Allow eg reaction time for absolute uncertainty	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Idea that each calculated value of T is actually $0.9T$ OR g becomes $\frac{g}{0.9^2}$ ✓ $= 12.1 \text{ (m s}^{-2}\text{)} \checkmark$	mp1 can be awarded for seeing $0.9T$	2	AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
11.1	Two waves travelling in opposite directions superpose ✓ Waves meet in phase / with zero phase difference ✓ Constructively interfere / produce maximum amplitude ✓	Maximum 2 marks if waves are described as 'coherent' If neither MP2 nor MP3 awarded, condone 1 mark for BOTH: Waves meet in antiphase / with π rad phase difference AND Destructively interfere / produce minimum amplitude Do not allow 'maximum displacement' for maximum amplitude in mp3	3	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
11.2	Use of $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$ OR $\mu = 0.0015 \text{ kg m}^{-1}$ ✓ 73 (N) ✓		2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.3	$(\frac{L}{4} \text{ leading to } 0.21 \text{ (m)})$ OR $(\frac{3L}{4} \text{ leading to } 0.63 \text{ (m)}) \checkmark$		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.4	Tension increases AND reference to $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \checkmark$ Tension increases by a (much) larger factor than the length $\checkmark$	Allow correct statement of proportionality between frequency and tension Allow ratio of (root) tension to (root) length increases For 'factor', allow words such as ratio, proportion etc	2	AO2
Total			8	

Question	Key	Answer	AO
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17	c	$0.91f$	A01
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18	B	0.80 m	AO2
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20	C	be coherent with each other.	AO2
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